

Wahyu Hendro Setyawan

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SUMMARY

Data engineer with 15 years owning large-scale operational and regulatory data at Indonesia's national electric utility (90M customers), now completing an M.S. in Data Science at Texas A&M (Dec 2026). Builds Python and SQL ETL/ELT pipelines, standardized data models, and data-quality and governance controls over multi-year transactional datasets. Recent work spans a genomic-selection pipeline over 1.07M field plots and agentic LLM systems over structured grid data.

SKILLS

Languages & Query: SQL, Python, JavaScript (D3.js, Next.js), C/C++, Bash

Data Stores: PostgreSQL, Cassandra, Redis, NoSQL, Parquet, Oracle ERP, Firebase

Data Engineering: ETL/ELT pipeline design, data modeling, schema design & standardization, data quality & validation, anomaly detection, data governance & privacy controls, feature engineering, Databricks, Docker, Git, HPC/parallel computing (CUDA, MPI)

Analytics & ML: pandas, NumPy, scikit-learn, statistical modeling, time-series analysis, data visualization, PyTorch, TensorFlow, model distillation, LLMs, RAG, multi-agent orchestration (MCP)

EXPERIENCE

Macroeconomic Data Analyst · Indonesian Ministry of State-Owned Enterprises, Jakarta Jan 2024 – Dec 2024

- Built Python and SQL ETL pipelines consolidating revenue and asset data from **65 state-owned enterprises and 1,000+ subsidiaries (USD 670B in assets)** into a single standardized reporting model, replacing inconsistent per-entity submissions.
- Designed the shared schema and validation rules that made cross-entity reporting comparable, resolving conflicting entity identifiers and unit conventions across sources.
- Modeled commodity price, national stock, and harvest-forecast time series to quantify supply risk; the analysis informed a policy decision that secured a **6-month national strategic reserve**.

Stakeholder Management Specialist · PLN (Indonesian State Electricity Company), Jakarta 2021 – Aug 2025

- Designed structured data-sharing and reporting workflows across **81 institutions**, defining the interchange formats, ownership, and turnaround expectations that removed recurring data-handoff bottlenecks.
- Acted as the data interface between engineering, regulatory, and government stakeholders, translating analytical findings into decisions for non-technical audiences.

Senior Regulatory Affairs Specialist · PLN, Jakarta Sep 2019 – 2021

- Owned data-driven regulatory monitoring for transmission and generation assets, translating **30+ regulations** and structured legal datasets into machine-checkable compliance rules applied across the asset base.

Chairperson, Procurement Committee · PLN Transmission & Substation Projects, Depok Sep 2016 – Sep 2019

- Audited structured procurement datasets in Oracle ERP covering **USD 300M+ in tenders, 200+ contractors, and 100+ executed contracts**, maintaining 100% data compliance under external audit.

Project Site Coordinator · System Validation Engineer · Commissioning Engineer · PLN, Indonesia Aug 2010 – Sep 2016

- Reconciled multi-source field progress data across **30+ concurrent grid projects (USD 60M)**; validated 1,000+ HV/EHV drawings and calculations. Authored a vendor-agnostic substation-automation data standard adopted nationally, replacing manufacturer-specific formats.

PROJECTS

BreedScope — genomic selection pipeline · 3rd place, 2026 AgTech Hackathon (Texas A&M × Bayer) Sep 2026

- Nine-stage Python pipeline over **1,072,276 field plots** ranking 15,968 untested maize lines for ~1,500 plot slots: ID reconciliation and contamination flagging, RR-BLUP breeding values from 2,911 SNP markers, forward-year validation,

factor-analytic GxE, and per-site advancement plans.

- Parameterized by marker rather than genome so X'X and X'y accumulate incrementally — **748 MB flat vs. ~63 GB** for a genomic relationship matrix over 125,495 lines; full run in 2 min 33 s.
- Largest accuracy gain (p 0.132 → 0.167) came from finding **nine site-years that pooled separate trials incorrectly** — a data-quality fix that beat every modeling change attempted. Shipped with a D3.js/Next.js decision dashboard on Firebase. github.com/hendrosetyawan/breedscope

X-GridAgent — on-policy distillation of an agentic LLM for power-grid analysis · Texas A&M Jan – May 2026

- Compressed a multi-agent GPT-class pipeline into a local 1B-parameter model at near-parity on power-flow, OPF, and contingency queries; orchestrated retrieval, SQL-generation, and validation agents over structured grid datasets via MCP tool servers and schema-adaptive RAG, with automated query quality checks.

EDUCATION

M.S. Data Science · Texas A&M University, College Station, TX Aug 2025 – Dec 2026 (expected)

B.S. Electrical Engineering · Institut Teknologi Bandung (ITB), Indonesia Aug 2005 – Oct 2010

CERTIFICATIONS

Model Context Protocol: Advanced Topics (Anthropic) · Building RAG Agents with LLMs (NVIDIA) · Redis for AI · SAP Cloud Platform Integration · CompTIA Data+ (beta tester)